Computing Voronoi Diagrams

1. Voronoi Diagrams
   - complexity and characterization
   - computing with CGAL
2. A Sweep Line Construction
   - the beach line
   - site events
   - circle events
3. A Sweep Line Algorithm
   - data structures
   - formulation of the algorithm

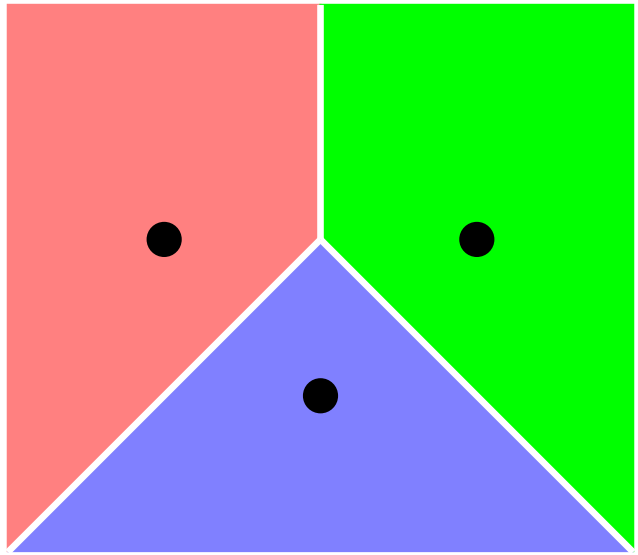
MCS 481 Lecture 22
Computational Geometry
Jan Verschelde, 14 October 2026

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complexity of the Voronoi diagram

For the post office problem, a Voronoi diagram partitions the plane in convex regions of all points closest to the same post office.

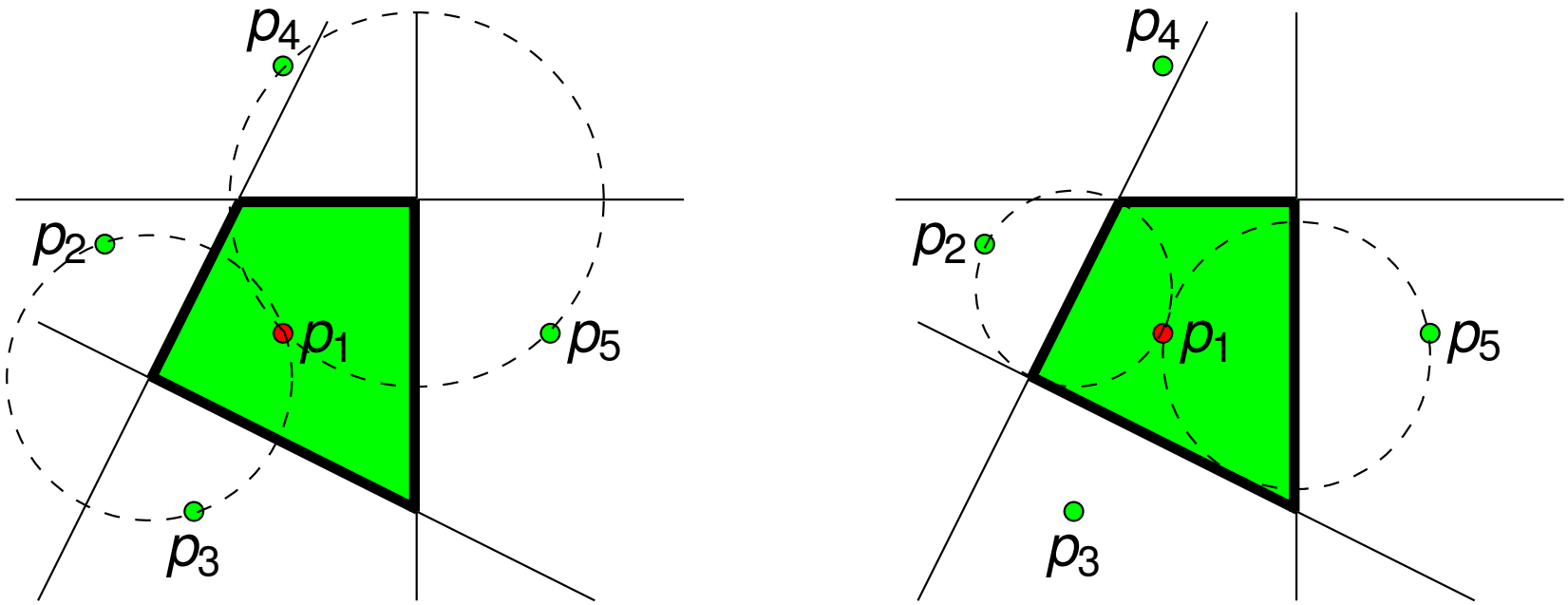


Theorem (complexity of a Voronoi diagram)

Let $\text{Vor}(P)$ *be a Voronoi diagram for* n *points,* $n \geq 3$.

1. *The number of vertices of* $\text{Vor}(P)$ *is at most* $2n - 5$.
2. *The number of edges of* $\text{Vor}(P)$ *is at most* $3n - 6$.

characterization of vertices and edges



Theorem (characterization of vertices and edges)

Let $\mathrm{Vor}(P)$ *be the Voronoi diagram of the set* P.

1. q *is a vertex of* $\mathrm{Vor}(P)$
   $\Leftrightarrow$ $C_P(q)$ *contains 3 or more points of* P *on its boundary.*
2. e *is an edge of* $\mathrm{Vor}(P) \Leftrightarrow$ e *is on the bisector between* p_i *and* p_j *and there is a* $q \in e$ *so* $C_P(q)$ *has only* p_i *and* p_j *on its boundary.*

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computing with CGAL

We use the Python bindings to CGAL.

On a simple set of sites:

$$\{(0,0),(100,0),(100,100),(0,100),(200,100),(100,200)\}.$$

The data is stored in a half edge structure.

To print the cells of the diagram, query points are used.

See the posted Jupyter notebook.

Exercise 1: Write code with matplotlib to plot all vertices and edges of the computed Voronoi diagram for this simple set of sites.

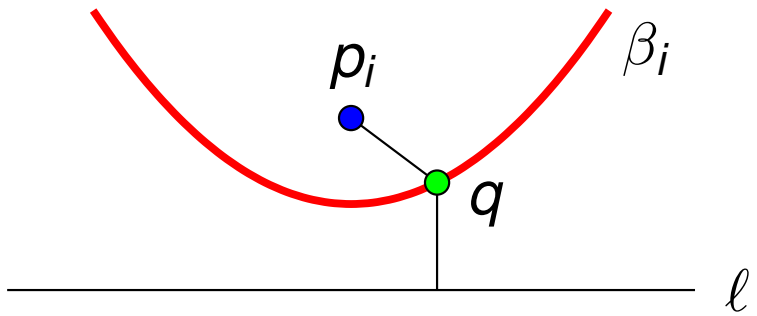
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a sweep line algorithm

The points in P are sorted on height in descending order:
points with highest y-coordinate precede the lower points.

Given is the point p_i and the line ℓ ,
all points q at equal distance from p_i and ℓ define
the parabola $\beta_i = \{\ q \in \mathbb{R}^2 \mid \text{dist}(q, p_i) = \text{dist}(q, \ell)\ \}$.

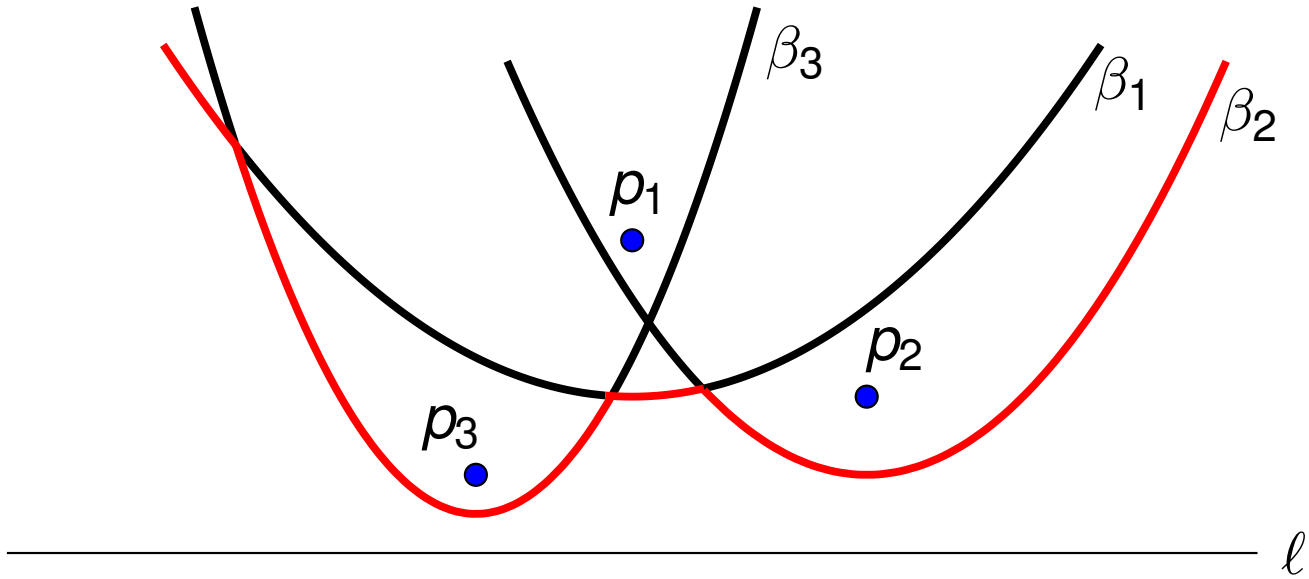


For every site p_i, we have the parabola β_i above the sweep line ℓ.

All points above β_i lie closer to p_i than to ℓ
$\Rightarrow$ everything of Vor(P) above β_i will not change as ℓ moves down.

the beach line

For the sites p_1, p_2, and p_3, we have parabolas β_1, β_2, and β_3:

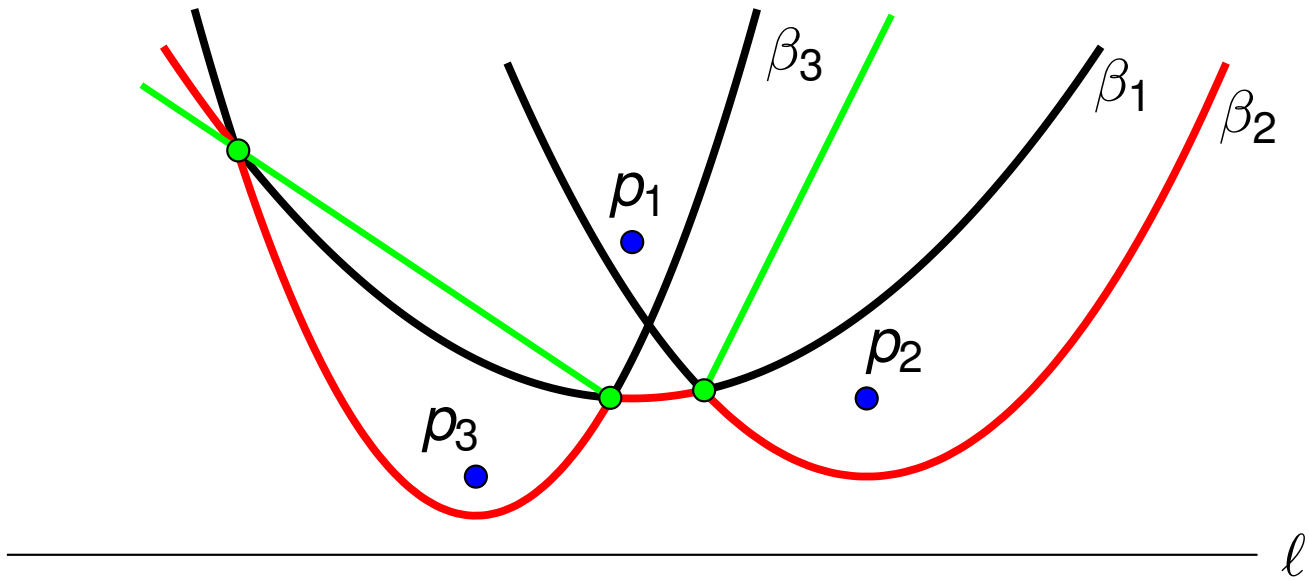


Everything of Vor(P) above β_i will not change as ℓ moves down.

Definition (the beach line)

The sequence of parabolic arcs that separates the points closer to any site than to ℓ is *the beach line*.

two observations about the beach line



1. The beach line is x-monotone: every vertical line intersects the beach line in exactly one point.
2. Breakpoints on the beach line, intersections between two adjacent parabolic arcs, lie on edges of the Voronoi diagram.

Exercise 2: Explain the observations by contradiction.

Computing Voronoi Diagrams

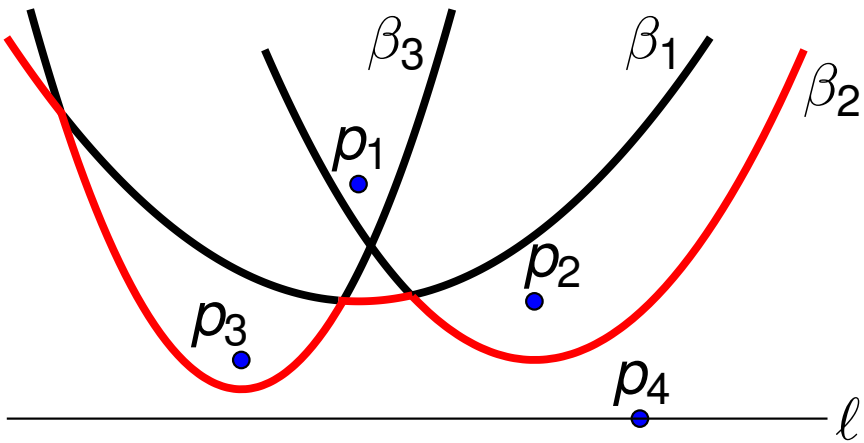
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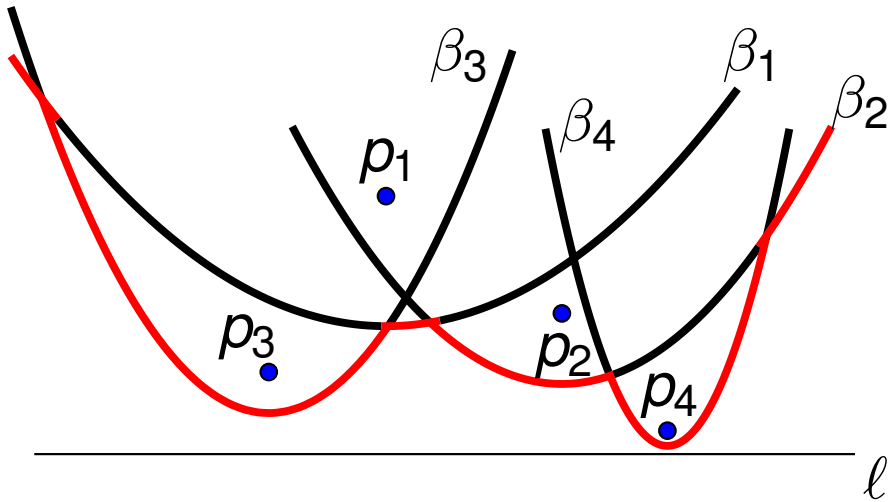
events during the sweep

As the sweep line ℓ moves down, the beach line may change.

1. A new parabolic arc appears.
2. A parabolic arc disappears.

If a new site appears on the sweep line,
then we have a new parabolic arc on the beach line:





Definition (site event)

The event where the sweep line meets a new site is *a site event*.

only through a site event appears a new arc

Lemma (new parabolic arc through site event)

The only way in which a new parabolic arc can appear on the beach line is through a site event.

Proof. By contradiction. Suppose an already existing parabola β_j defined by p_j breaks through the beach line.

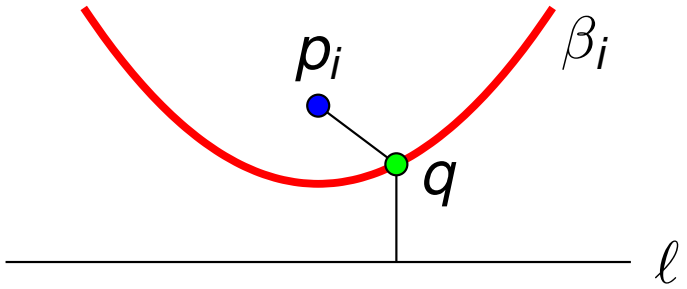
There are two ways β_j can break through the beach line:

1. β_j breaks through the middle of arc β_i, or
2. β_j appears in between two arcs.

To handle the first case, consider the equation for the parabola, to compute the intersection point where β_i and β_j touch.

the equation for the parabola

To write the equation for $\beta_i = \{\ q \in \mathbb{R}^2 \mid \text{dist}(q, p_i) = \text{dist}(q, \ell)\ \}$
we select coordinates: $q = (x, y)$, $p_i = (p_{i,x}, p_{i,y})$, $\ell : y = y_\ell$.



Points on β_i are equidistant to the focus p_i and the directrix ℓ:

$$\begin{aligned}
\text{dist}(q, p_i) &= \text{dist}(q, \ell) \\
&\Leftrightarrow (x - p_{i,x})^2 + (y - p_{i,y})^2 = (y - y_\ell)^2 \\
&\Leftrightarrow (x - p_{i,x})^2 + \cancel{y^2} - 2yp_{i,y} + p_{i,y}^2 = \cancel{y^2} - 2yy_\ell + y_\ell^2
\end{aligned}$$

We solve for y and obtain $y = \dfrac{1}{2(p_{i,y} - y_\ell)} \left((x - p_{i,x})^2 + p_{i,y}^2 - y_\ell^2 \right)$.

computing the intersection point

The equation for parabola β_i:

$$y = \frac{1}{2(p_{i,y} - y_\ell)} \left((x - p_{i,x})^2 + p_{i,y}^2 - y_\ell^2 \right).$$

The equation for parabola β_j:

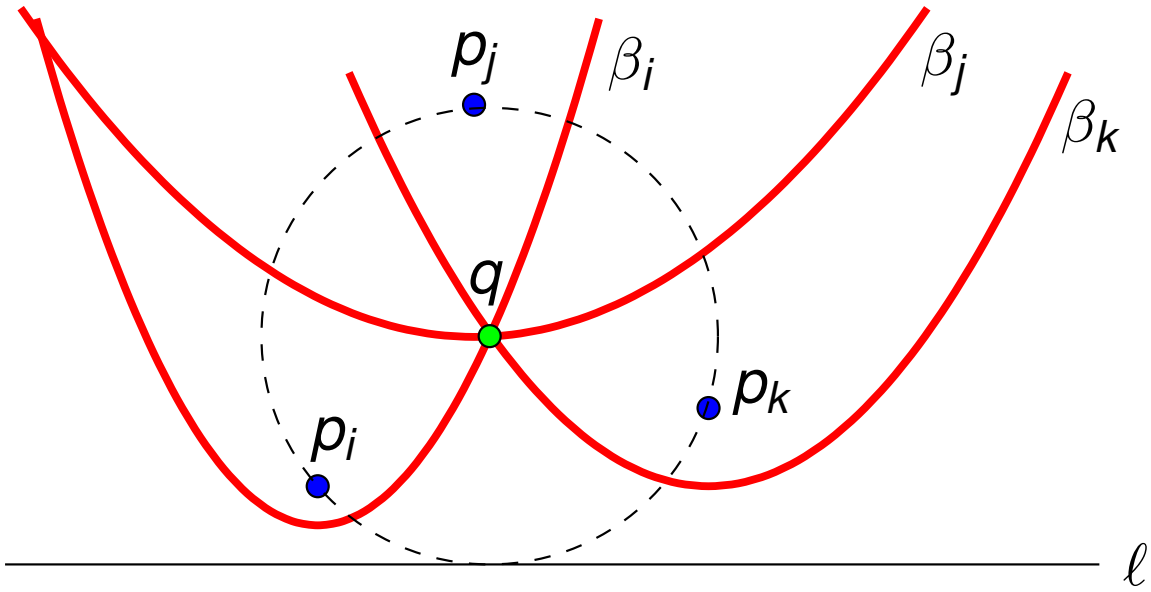
$$y = \frac{1}{2(p_{j,y} - y_\ell)} \left((x - p_{j,x})^2 + p_{j,y}^2 - y_\ell^2 \right).$$

Eliminating y leads to one quadratic equation in x.

This quadratic equation has two solutions for x,
but the beach line is x-monotone, contradicting the first case.

a new parabolic arc in between two arcs

A new arc β_j appears between two arcs, between β_i and β_k:

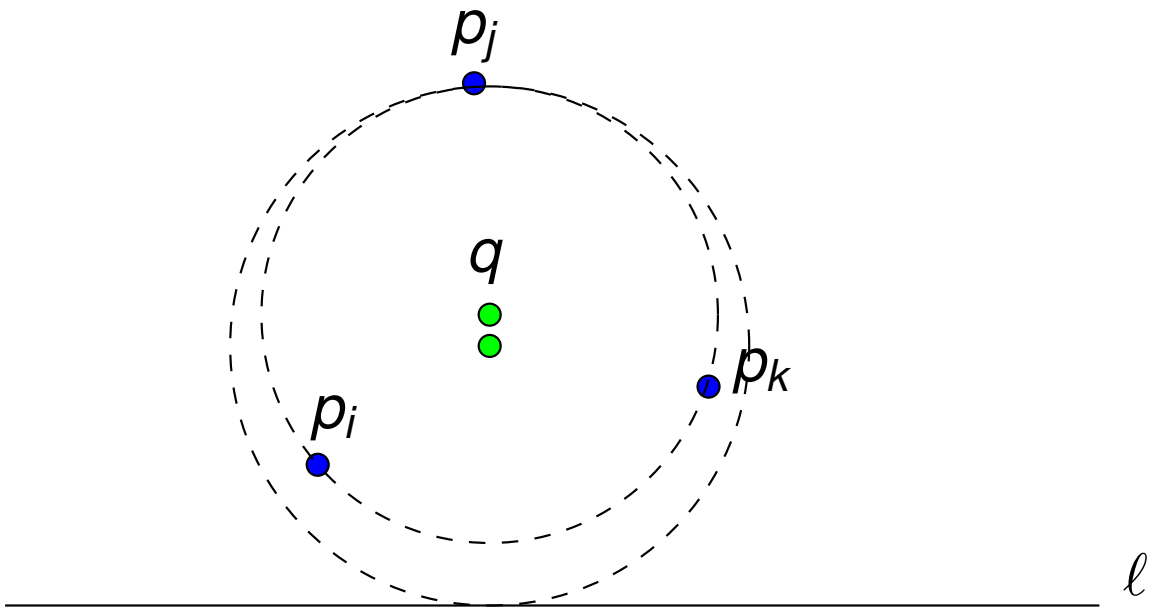


The intersection point q is the center of $C_P(q)$,
with the points p_i, p_j, and p_k on the boundary of $C_P(q)$,
also tangent to the sweep line ℓ.

Three sites on the boundary of $C_P(q) \Leftrightarrow q$ is a vertex of Vor(P).

the sweep line comes down

As the sweep line comes down, while keeping the circle $C_P(q)$ tangent, keeping p_j on its boundary will enlarge so that p_i and/or p_k will be part of the interior of $C_P(q)$.



Therefore, β_j cannot appear on the beach line as ℓ moves down, because either p_i or p_k will be closer to ℓ than p_j.

Q.E.D.

on the number of arcs on the beach line

Lemma (new parabolic arc through site event)

The only way in which a new parabolic arc can appear on the beach line is through a site event.

The only way for a new parabolic arc to appear is to split through an existing arc at a site event.

Because there are n points on input, there can be at most $n - 1$ splits after the first site.

Therefore, we have the following bound on the number of arcs:

Corollary (number of arcs on the beach line)

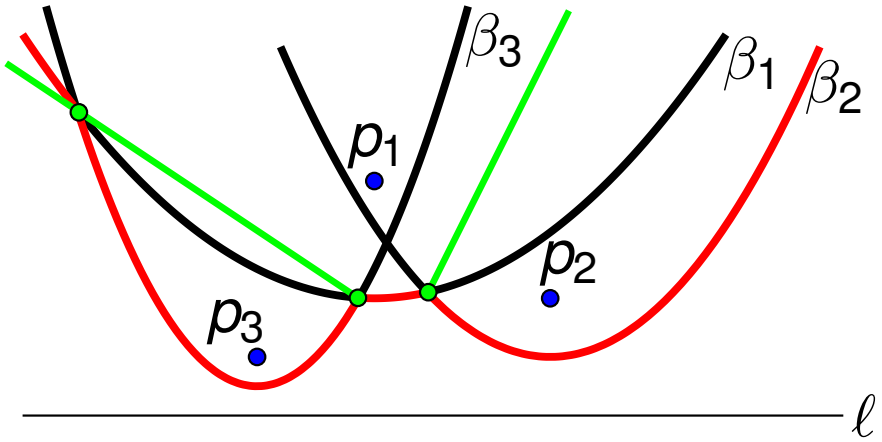
The number of arcs on the beach line is bounded by $2n - 1$.

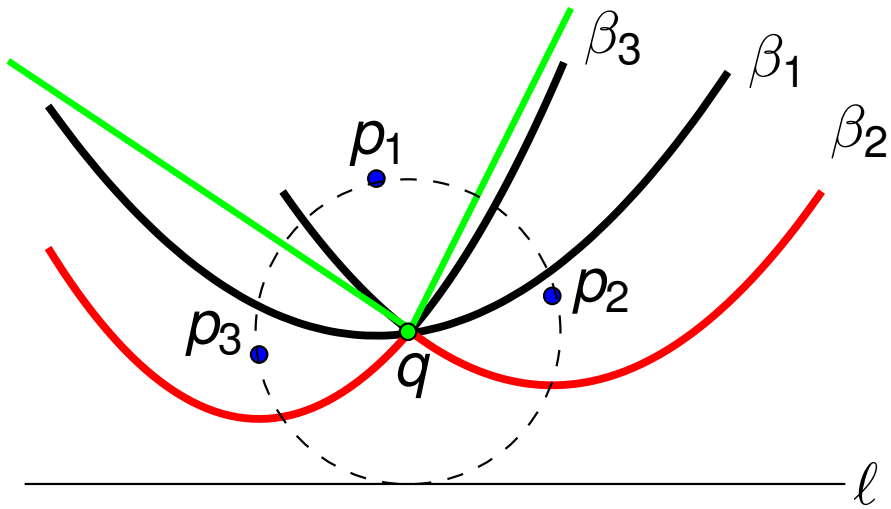
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a parabolic arc disappears

The second type of event is when a parabolic arc disappears.





At the moment the arc β_1 disappears, β_1, β_2, and β_3 pass through a vertex q equidistant to p_1, p_2, and p_3.

Definition (circle event)

Where the sweep line reaches the lowest point on a circle through 3 sites defining consecutive arcs on the beach line is a *circle event*.

two types of events during the sweep

We just derived the following:

Lemma (disappearing parabolic arc through circle event)

The only way in which an existing arc can disappear from the beach line is through a circle event.

Earlier we proved the following:

Lemma (new parabolic arc through site event)

The only way in which a new parabolic arc can appear on the beach line is through a site event.

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data structures

We have three data structures:

1. A doubly connected edge list $\mathcal{D}$ stores Vor(P).
   For a true subdivision, we need a bounding box.
2. The beach line is represented by a balanced binary search tree $\mathcal{T}$: adjacent leaves are adjacent arcs. Internal nodes just above arcs represent breakpoints between arcs.
3. A priority queue Q stores site and circle events. A circle event is defined by a triple of arcs and the lowest point of $C_P(q)$.

Circle events are not known in advance.

Exercise 3: Consider $P = \{ (2,3), (0,4), (1,1), (2,6), (5,3) \}$.
Construct a bounding box and describe a doubly connected edge list to store Vor(P).

vertices and circle events

A vertex q of Vor(P) is characterized by a largest empty circle $C_P(q)$ which has at least three points on its boundary.

Lemma (vertices and circle events)

Every Voronoi vertex is detected by a circle event.

At a circle event, an arc disappears from the beach line.

Consider the neighbors β_i and β_k of the disappearing arc β_j, defined respectively by the sites p_i, p_k, and p_j.

- Let $C(p_i, p_j, p_k)$ be the circle that passes through the sites.
- The circle event happens when the lowest point of $C(p_i, p_j, p_k)$ touches the sweep line ℓ.

For every vertex q, there is thus a circle $C(p_i, p_j, p_k)$, which corresponds to the largest empty circle $C_P(q)$, constructed at the circle event. Q.E.D.

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formulation of the algorithm

Algorithm VORONOIDIAGRAM(P)

Input: n sites in the set P.
Output: Vor(P) stored in a doubly connected edge list $\mathcal{D}$.

1. Initialize Q with P; $\mathcal{T} = \emptyset$; $\mathcal{D} = \emptyset$
2. while $Q \neq \emptyset$ do
3. pop next event e from Q
4. if e is a site event
5. then HANDLESITEEVENT(e), e is a site p_i
6. else HANDLECIRCLEEVENT(γ), γ is leaf of $\mathcal{T}$
7. compute bounding box R for half line edges
8. update $\mathcal{D}$ with R

handling a site event

Algorithm HANDLESITEEVENT($Q, \mathcal{T}, \mathcal{D}, p_i$)

Input: event queue Q, tree $\mathcal{T}$, DCEL $\mathcal{D}$, site p_i.

1. if $\mathcal{T} = \emptyset$ then
2. insert p_i in $\mathcal{T}$

else

3. let α be the arc in $\mathcal{T}$ above p_i
4. if α has a pointer to a circle event e in Q, then delete e from Q
5. replace leaf in $\mathcal{T}$ for α with subtree of 3 leaves
6. rebalance $\mathcal{T}$ if necessary
7. create new half edge records in $\mathcal{D}$ for edges separating cells
8. for all triple consecutive arcs that contain p_i as breakpoints
9. add circle events to Q

handling a circle event

Algorithm HANDLECIRCLEEVENT($Q, \mathcal{T}, \mathcal{D}, \gamma$)

Input: event queue Q, tree $\mathcal{T}$, DCEL $\mathcal{D}$, leaf γ of $\mathcal{T}$.

1. update $\mathcal{T}$:
   1. delete leaf γ which represents the disappearing arc from $\mathcal{T}$
   2. update breakpoints at internal nodes, rebalance $\mathcal{T}$ if necessary
   3. delete circle events involving α
2. update $\mathcal{D}$:
   1. insert new vertex record for center q of $C_P(q)$
   2. create two half edge records for new breakpoints
   3. set pointers between half edges, attach new half edges at new q
3. update Q:
   1. search for new circle events created by disappearing arc α
   2. insert these new circle events into Q

summary and exercises

The site and circle events in a sweep line algorithm construct all vertices and edges of a Voronoi diagram.

We covered section 7.2 in the textbook.

Consider the following activities, listed below.

1. Write the solutions to exercises 1, 2, and 3.
2. Consult the CGAL documentation and example code on Voronoi diagrams.
3. Consider the exercises 4, 6, 8 in the textbook.

Steven Fortune: **A Sweepline Algorithm for Voronoi Diagrams.** *Algorithmica*, Volume 2, pages 153–174, 1987.